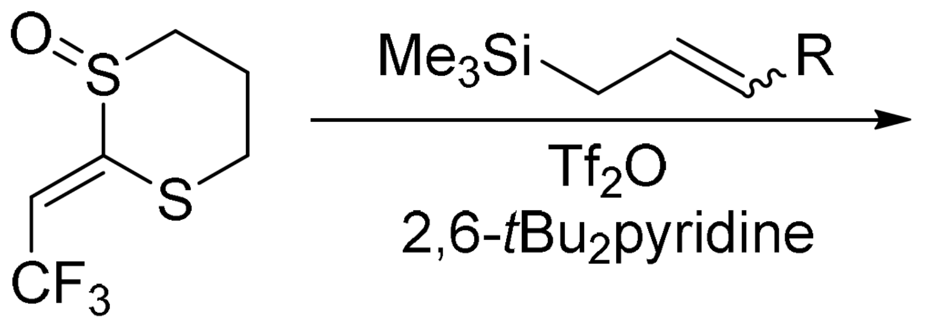
Question

In the Pummerer rearrangement, when the substrate sulfoxide is replaced with an alkenyl sulfoxide $\mathbf{A}$, the original Pummerer rearrangement process will not occur. Alkenyl sulfoxide $\mathbf{A}$ exhibits different reaction properties when reacting with alkenes substituted with different R groups (as shown in the figure below).



O=S(CCCS/1)C1=C\C(F)(F)F.C[Si](C)(C/C=C/[R])C>Tf_2O, $2,6-t-Bu_2pyridine$>[X]. [X] represents the reaction product.

When the substrates are different, the reaction product $\mathbf{X}$ may be $\mathbf{B, C, D}$

When $\mathrm{R} = \mathrm{H}$, the molecular formula of the main reaction product $\mathbf{B}$ is $\mathrm{C_9H_{11}F_3S_2}$; when $\mathrm{R} = \mathrm{C_7H_{15}}$, only one product $\mathbf{C}$ was obtained, and it is known that the formation of $\mathbf{C}$ involves a pericyclic reaction.

Phenol reacts with alkenyl sulfoxide $\mathbf{A}$ under the action of $\mathrm{Tf_2O}$ to obtain the spirocyclic compound $\mathbf{D}$.

Which of the following statements about $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ are correct:

1. There are two conjugated carbon-carbon double bonds in $\mathbf{B}$

2. When the substrate is isotopically labeled, i.e., when $\mathrm{R} = \mathrm{D}$, the deuterium atom in the main reaction product $\mathbf{B}'$ is separated from the nearest sulfur atom by three carbon atoms

3. A [3,3]-σ migration involving an oxygen atom occurred in the process of generating $\mathbf{C}$

4. The R group (i.e., $\mathrm{C_7H_{15}}$) in $\mathbf{C}$ is connected to a primary carbon

5. There are two conjugated carbon-carbon double bonds in **C**

6. **D** is chiral

A. 1,2,4,6

B. 1,2,3,4,6

C. 1,2,3,4,5,6

D. 1,2,6

E. 1,3,5,6

F. 2,4,6

G. 2,4

H. 1,2,3,5,6

I. 1,5,6

J. 3,6

K. 1,2,3,6

L. 2,3,4,5,6

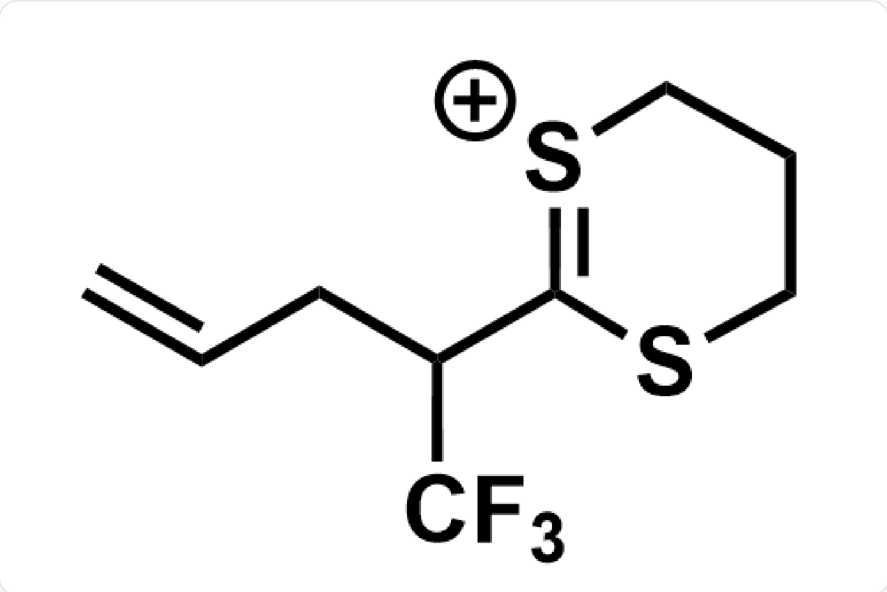
M. All of the above options are incorrect or the answer is incomplete.

Answer

Correct Answer: F

Detailed Explanation

The silyl group at the β position of an alkene can enhance the nucleophilicity of the carbon-carbon double bond through the hyperconjugation effect of the carbon-silicon single bond on the carbon-carbon double bond. Alkenyl sulfoxide $\mathbf{A}$ first generates a sulfonium ion under the action of trifluoroacetic anhydride, and then is attacked by the substrate alkene at the terminal carbon-carbon double bond conjugated to the sulfonium ion via an $\mathrm{S_N2'}$ mechanism, leaving the silyl group to obtain a carbon-carbon double bond at the end of the chain, as shown in the following structure:



C=CCC(C1=[S+]CCCS1)C(F)(F)F

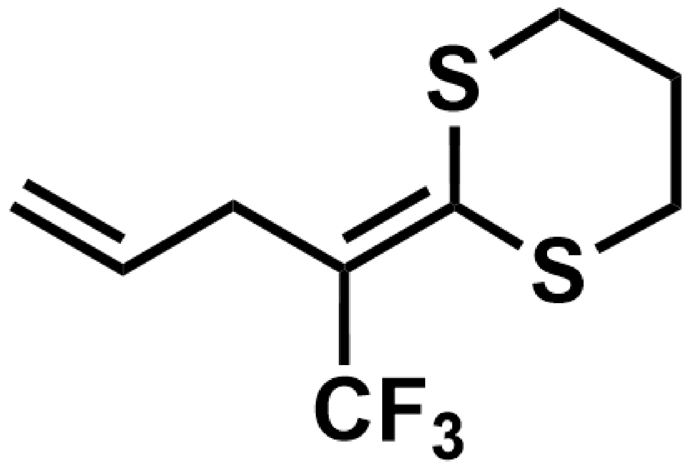
Finally, a proton is eliminated to form a carbon-carbon double bond conjugated with two sulfur atoms, and the final structure of $\mathbf{B}$ is as follows:

CHECKPOINT 1 PTS

The sulfonium ion generated from alkenyl sulfoxide **A** is attacked by the substrate alkene via an S_N2' mechanism

CHECKPOINT 1 PTS

Elimination of a proton forms a carbon-carbon double bond conjugated with two sulfur atoms



C=CC/C(C(F)(F)F)=C1SCCCS\1

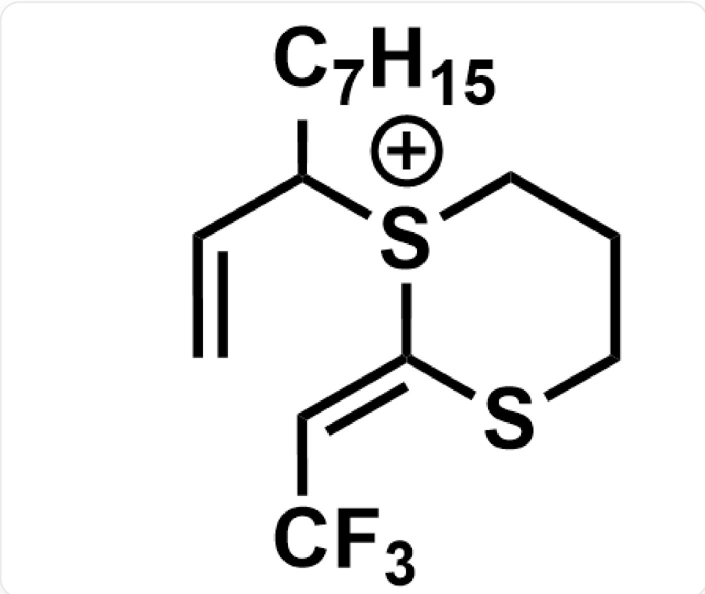
CHECKPOINT 1 PTS

The structure of **B** is C=CC/C(C(F)(F)F)=C1SCCCS\1

The two carbon-carbon double bonds in $\mathbf{B}$ are not conjugated, statement 1 is incorrect.

When $\mathrm{R} = \mathrm{D}$, according to the reaction mechanism, the deuterium atom in the main product $\mathbf{B}'$ should be separated from the nearest sulfur atom by three carbon atoms, so statement 2 is correct.

When $\mathrm{R} = \mathrm{C_7H_{15}}$, the steric hindrance is large, and the generated sulfonium ion is directly nucleophilically attacked by the substrate alkene to obtain a sulfonium ion connected to three carbons. At the same time, the silyl group leaves to obtain a carbon-carbon double bond at the end of the chain, as shown in the following structure:



C=CC([S+](CCCS/1)C1=C\C(F)(F)F)CCCCCCC

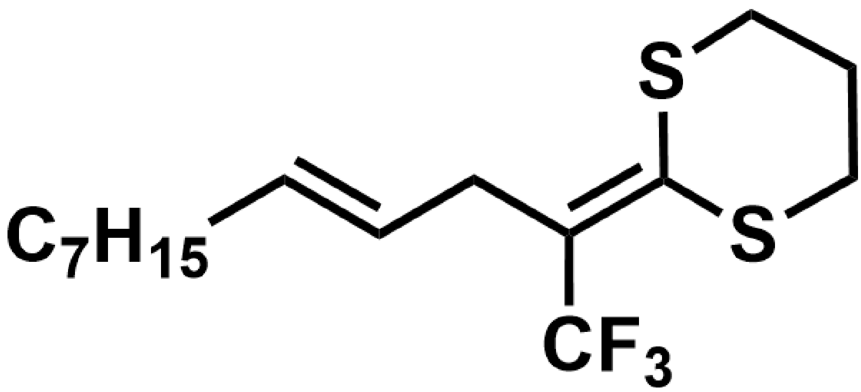
[3,3]-σ migration occurs, and a proton is eliminated to form a carbon-carbon double bond conjugated with two sulfur atoms to obtain $\mathbf{C}$, as shown in the following structure:

CHECKPOINT 1 PTS

When $\mathrm{R} = \mathrm{C_7H_{15}}$, the generated sulfonium ion is directly nucleophilically attacked by the substrate alkene

CHECKPOINT 1 PTS

A [3,3]-σ migration occurs and a proton is eliminated to form a carbon-carbon double bond conjugated with two sulfur atoms



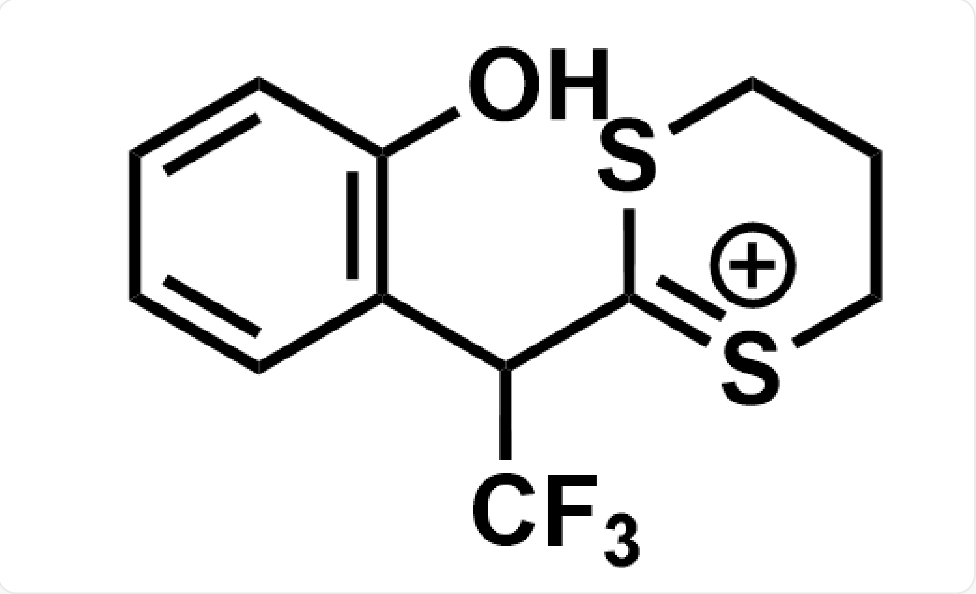
FC(/C(C/C=C/CCCCCCC)=C1SCCCS/1)(F)F

CHECKPOINT 1 PTS

The structure of $\mathbf{C}$ is FC(/C(C/C=C/CCCCCCC)=C1SCCCS/1

Therefore, the oxygen atom is not involved in the [3,3]-σ migration, C_7H_{15} is connected to a primary carbon, and the two carbon-carbon double bonds in $\mathbf{C}$ are also not conjugated, so statement 4 is correct, and statements 3 and 5 are incorrect.

When the substrate is replaced with phenol, similarly to the process of generating $\mathbf{B}$, the ortho position of the electron-rich hydroxyl group of phenol attacks the terminal carbon-carbon double bond conjugated with the sulfonium ion via an $\mathrm{S_N2'}$ mechanism, and then a proton is eliminated to restore aromaticity; the trifluoroacetate anion leaves to obtain a sulfonium ion with a carbon-sulfur double bond-like structure, as shown in the following structure:



OC1=C(C(C2=[S+]CCCS2)C(F)(F)F)C=CC=C1

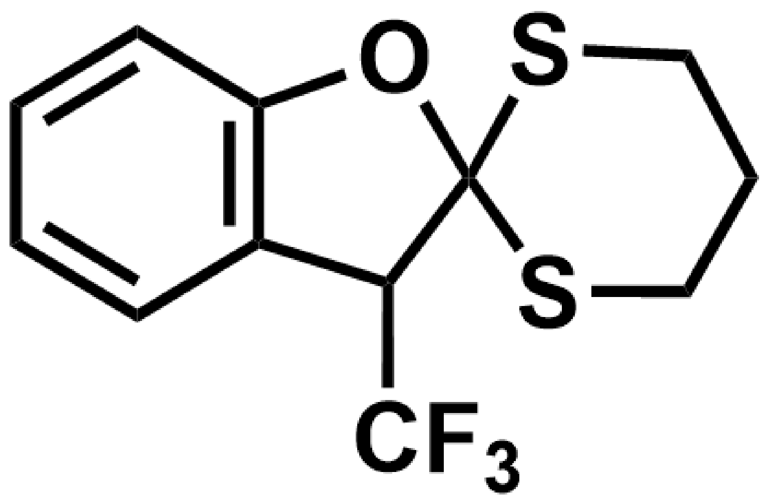
However, since there is no hydrogen in the ortho position at this time to eliminate, the phenolic hydroxyl group attacks the carbon-sulfur double bond to finally obtain the five-membered spirocyclic structure **D**, as shown in the following structure:

CHECKPOINT 1 PTS

The ortho position of the electron-rich hydroxyl group of phenol attacks the terminal carbon-carbon double bond conjugated with the sulfonium ion via an S_N2' mechanism

CHECKPOINT 1 PTS

The phenolic hydroxyl group attacks the carbocation to finally obtain the five-membered spirocyclic structure



FC(C1C2(OC3=C1C=CC=C3)SCCCS2)(F)F

CHECKPOINT 1 PTS

The structure of $\mathbf{D}$ is FC(C1C2(OC3=C1C=CC=C3)SCCCS2

The benzylic position of the benzene ring in $\mathbf{D}$ is connected to four different groups. At the same time, there is no mirror plane, center of symmetry, or four-fold improper rotation axis in $\mathbf{D}$, so $\mathbf{D}$ is chiral, and statement 6 is correct.

The correct answer is F